

LandTrendr-Based Wildfire Disturbance and Recovery Analysis

Project documentation for a GitHub portfolio repository

Case study: 2013 Rim Fire, Stanislaus National Forest, California

Project Snapshot

Study area	Rim Fire burn region, Tuolumne County, California, USA
Temporal coverage	Annual Landsat composites, 1984–2021
Input data	Landsat 5 TM, Landsat 7 ETM+, Landsat 8 OLI, plus DEM-derived slope
Core method	LandTrendr time-series segmentation in Google Earth Engine
Key outputs	Year of Detection (YOD), burn severity, recovery rate, and slope–recovery comparison

Project Overview

This document repackages the original academic assessment as project-facing documentation suitable for a GitHub repository. The project applies the LandTrendr algorithm to Landsat time series to identify wildfire disturbance, map burn severity, track post-fire vegetation recovery, and test whether recovery varies across slope classes.

- Map the main wildfire disturbance event and extract the Year of Detection (YOD).
- Quantify disturbance characteristics including magnitude, duration, and rate of change.
- Visualise post-fire recovery patterns using LandTrendr-derived NBR recovery metrics.
- Explore whether post-fire recovery differs across terrain slope classes.

Study Area and Data

The study focuses on the **2013 Rim Fire**, which burned within Stanislaus National Forest in Tuolumne County, California. The analysis uses annual Landsat composites from 10 June to 20 September each year to reduce seasonal noise and support consistent multi-year comparison. A DEM was used to derive slope for exploratory terrain analysis.

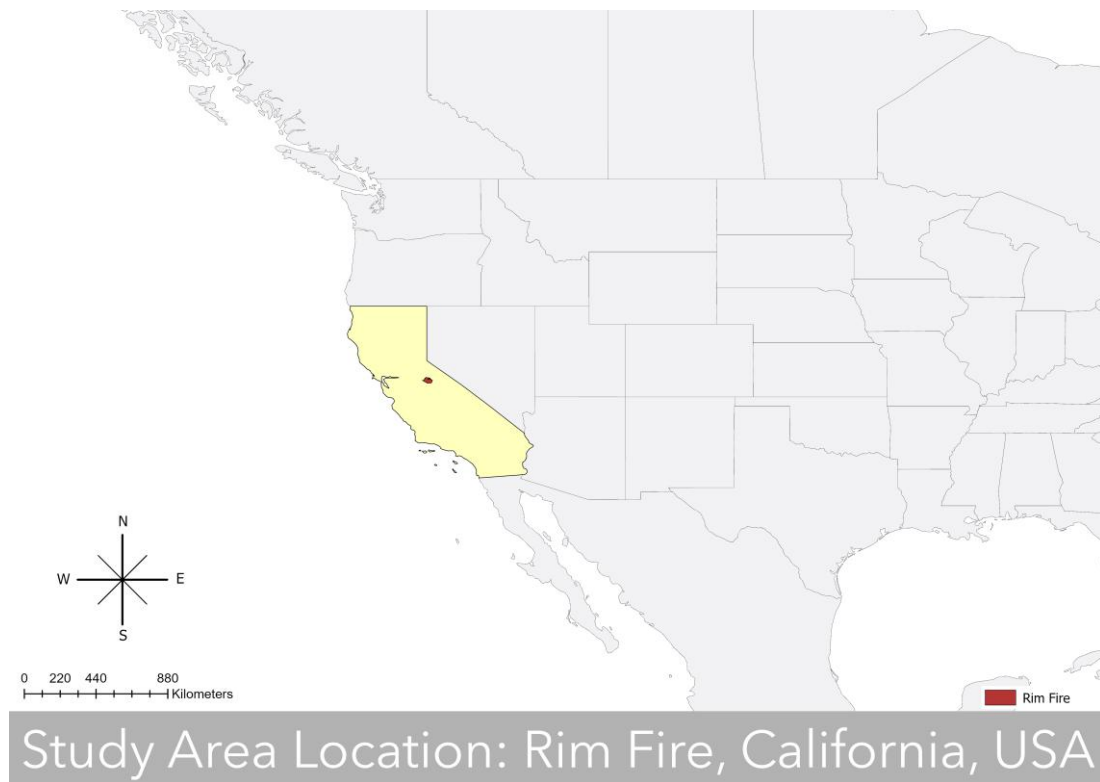


Figure 1. Study area location.

Analytical Workflow

The workflow combines remote-sensing pre-processing, LandTrendr temporal segmentation, disturbance filtering, recovery metric extraction, and terrain-based comparison.

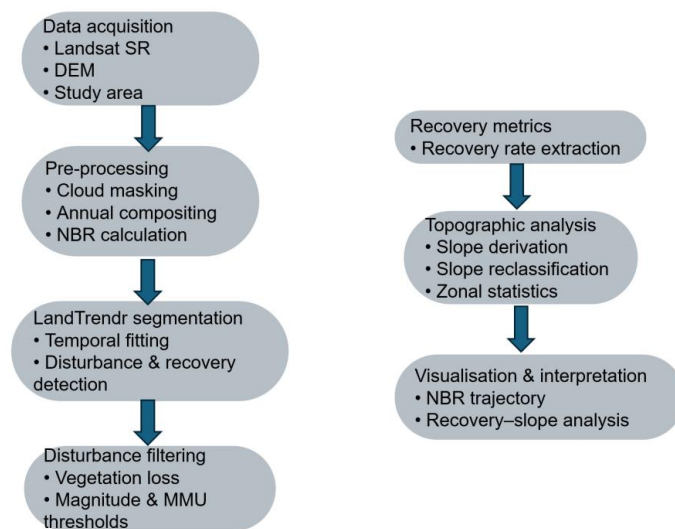


Figure 2. Simplified workflow used for the project.

Method Summary

LandTrendr parameterisation was configured to target vegetation loss associated with the Rim Fire while filtering out minor spectral fluctuations and isolated noise. *A weak negative slope-recovery relationship was explored descriptively rather than treated as a strong causal model.*

Parameter	Setting and rationale
Change direction	Delta = loss; focuses the analysis on canopy decline caused by wildfire.

Segment selection	Sort = greatest; extracts the strongest disturbance segment in the time series.
Target years	2011–2015; captures the immediate pre-fire, fire, and early post-fire period.
Magnitude threshold	> 500 NBR units; filters out minor changes and sensor/seasonal noise.
Pre-disturbance threshold	> 400; restricts disturbance detection to vegetated land.
Minimum Mapping Unit	11 pixels; removes isolated noise and improves spatial continuity.

Key Results

The project produced spatial and statistical outputs that describe when disturbance occurred, where burn severity was highest, how recovery varied across the landscape, and whether slope explained part of that recovery pattern.

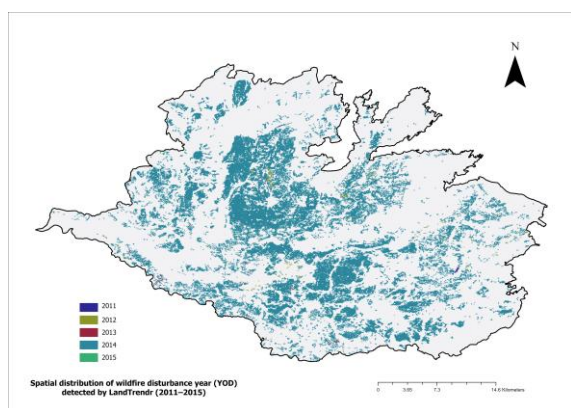


Figure 3. Year of Detection (YOD) map. Most detected disturbance pixels were assigned to 2014 in the annual composite series.

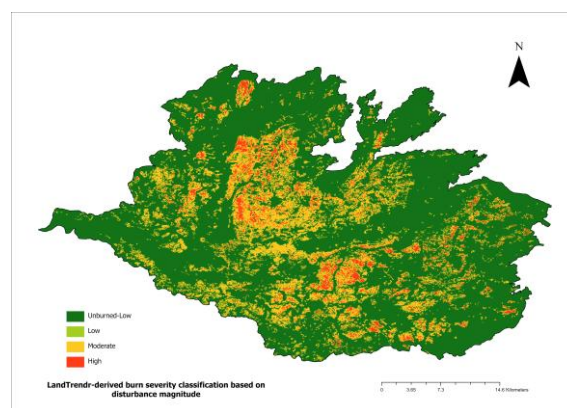


Figure 4. Burn severity classification derived from LandTrendr disturbance magnitude.

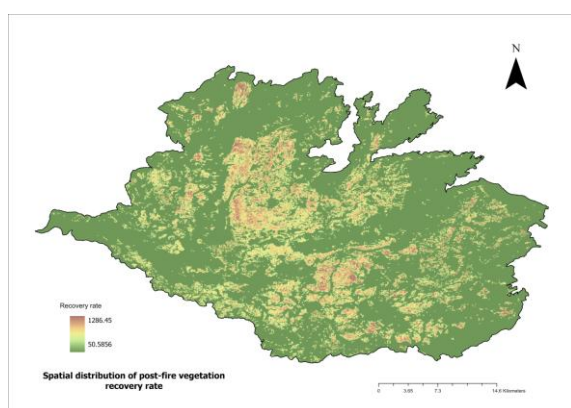


Figure 5. Spatial distribution of post-fire recovery rate.

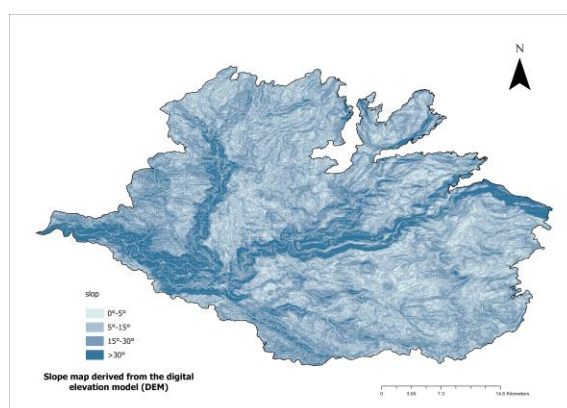


Figure 6. Slope of study area derived from DEM

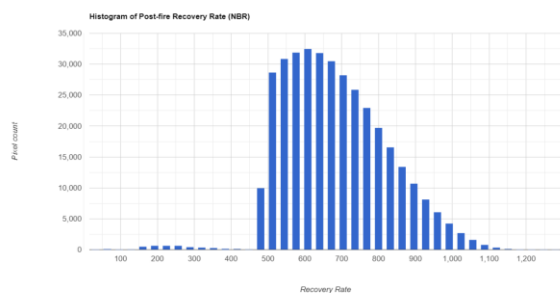


Figure 7. Histogram of post-fire recovery rate values.

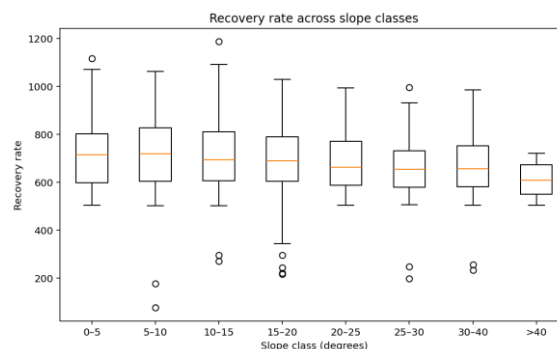


Figure 8. Box plot comparing recovery rate across slope classes.

Interpretation

- Disturbance pixels clustered spatially rather than being evenly distributed across the study area.
- The dominant disturbance year was 2014 in the annual LandTrendr output, consistent with the main Rim Fire signal captured in annual composites.
- Burn severity and post-fire recovery both showed strong spatial heterogeneity.
- Recovery tended to be lower on steeper slopes, but the relationship was weak and should be interpreted cautiously.

Project Limitations

- LandTrendr identifies spectral disturbance rather than wildfire alone, so other disturbance types may be difficult to separate completely.
- Annual composites simplify short-term dynamics and may smooth rapid post-fire change.
- Slope was analysed as an exploratory terrain factor only; post-fire recovery is also likely influenced by burn severity, vegetation type, light conditions, and human intervention.

Suggested Repository Contents

This project would sit naturally in a GitHub repository with a [README.md](#) plus folders for scripts, figures, and exported outputs. A simple structure could be:

```
project-root/
├── README.md
├── docs/
│   └── RimFire_Project_Documentation.docx
├── figures/
│   ├── study_area.png
│   ├── workflow.png
│   ├── yod.png
│   ├── burn_severity.png
│   ├── slope.tif
│   ├── recovery_slope_boxplot.png
│   ├── recovery_rate.png
│   ├── trajectory.png
│   └── recovery_histogram.png
└── scripts/
    └── gee_landtrendr_rimfire.js
```

References

- Cohen, W.B. et al. (2016). Forest disturbance across the conterminous United States from 1985–2012: The emerging dominance of forest decline. *Forest Ecology and Management*, 360, 242–252.
- Kennedy, R.E., Yang, Z. and Cohen, W.B. (2010). Detecting trends in forest disturbance and recovery using yearly Landsat time series: 1. LandTrendr—Temporal segmentation algorithms. *Remote Sensing of Environment*, 114(12), 2897–2910.